MAGIVOX AI

Aptitude & Reasoning - A to Z

Arithmetic | Percentages | Ratio | Time & Work | Logic | Practice Strategy

How to use this guide
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

Learning outcomes

- BODMAS
- Number Basics
- Percentages
- Ratio and Proportion
- Average
- Profit, Loss and Discount

Table of Contents

1. BODMAS
2. Number Basics
3. Percentages
4. Ratio and Proportion
5. Average
6. Profit, Loss and Discount
7. Simple and Compound Interest
8. Time and Work
9. Speed, Time and Distance
10. Number Series
11. Logical Reasoning
12. Data Interpretation
13. Test Strategy

1. BODMAS

What to learn

- Brackets
- Orders/powers
- Division and multiplication left-to-right
- Addition and subtraction left-to-right

Practice checkpoint

Explain brackets in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

2. Number Basics

What to learn

- Factors and multiples
- LCM/HCF
- Divisibility
- Remainders

Practice checkpoint

Explain factors and multiples in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

3. Percentages

What to learn

- Percentage of a number
- Increase/decrease
- Successive percentage change

Practice checkpoint

Explain percentage of a number in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

4. Ratio and Proportion

What to learn

- Simplifying ratio
- Sharing in a ratio
- Direct proportion

Practice checkpoint

Explain simplifying ratio in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

5. Average

What to learn

- Simple average
- Weighted average idea
- Missing value

Practice checkpoint

Explain simple average in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

6. Profit, Loss and Discount

What to learn

- Cost price/selling price
- Profit percent
- Discount
- Marked price

Practice checkpoint

Explain cost price/selling price in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

7. Simple and Compound Interest

What to learn

- Simple interest formula
- Compound growth concept

Practice checkpoint

Explain simple interest formula in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

8. Time and Work

What to learn

- Work rate
- Combined work
- Pipes/cistern analogy

Practice checkpoint

Explain work rate in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

9. Speed, Time and Distance

What to learn

- D=ST
- Relative speed
- Trains basics

Practice checkpoint

Explain d=st in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

10. Number Series

What to learn

- Difference patterns
- Multiplication patterns
- Alternating patterns

Practice checkpoint

Explain difference patterns in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

11. Logical Reasoning

What to learn

- Syllogisms
- Odd one out
- Analogy
- Coding-decoding basics

Practice checkpoint

Explain syllogisms in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

12. Data Interpretation

What to learn

- Tables
- Bar charts
- Percent change
- Approximation

Practice checkpoint

Explain tables in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

13. Test Strategy

What to learn

- Read question carefully
- Estimate
- Skip and return
- Track mistakes
- Timed practice

Practice checkpoint

Explain read question carefully in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

Worked Examples

BODMAS

```
18 - 6 / 2 + 3 = 18 - 3 + 3 = 18
```

Percentage

```
25% of 240 = 240 / 4 = 60
```

Ratio

```
3:5 total 64 -> 8 parts -> 8 each -> smaller = 24
```

Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.